

D - Greedy Customer

分值: 700 分

Problem Statement

You are given positive integers N, M and a sequence of positive integers $A = (A_1, A_2, \dots, A_N)$ of length N .

给定正整数 N, M , 以及一个长度为 N 的正整数序列 $A = (A_1, A_2, \dots, A_N)$ 。

For a positive integer k , define $f(k)$ as the answer to the following problem.

对于正整数 k , 定义 $f(k)$ 为下述问题的答案。

Takahashi has k yen. AtCoder Store has N items, and the i -th item costs A_i yen.

He performs the following action for $i = 1, 2, \dots, N$ in order.

- If the amount of money he currently has is at least A_i yen, he purchases the i -th item.

Find the total price of the items he purchased.

高桥有 k 日元。AtCoder 商店有 N 件商品, 第 i 件商品的价格为 A_i 日元。

他按顺序对 $i = 1, 2, \dots, N$ 执行以下操作:

- 如果他当前持有的钱不少于 A_i 日元, 就购买第 i 件商品。

求他购买的商品的总价格。

Find $\bigoplus_{1 \leq k \leq M} (k \times f(k))$. Here, $\bigoplus_{1 \leq k \leq M} (k \times f(k))$ is defined as the bitwise XOR of $1 \times f(1), 2 \times f(2), \dots, M \times f(M)$.

请求出 $\bigoplus_{1 \leq k \leq M} (k \times f(k))$ 。其中 $\bigoplus_{1 \leq k \leq M} (k \times f(k))$ 定义为 $1 \times f(1), 2 \times f(2), \dots, M \times f(M)$ 的按位 XOR。

You are given T test cases; solve each of them.

给定 T 组测试数据, 请你分别求解每组数据。

What is bitwise ?

什么是按位 XOR?

The bitwise XOR of non-negative integers A and B , $A \oplus B$, is defined as follows.

非负整数 A 和 B 的按位 XOR 结果记为 $A \oplus B$, 其定义如下。

- In the binary representation of $A \oplus B$, the digit at the 2^k ($k \geq 0$) place is 1 if exactly one of the digits at the 2^k place in the binary representations of A and B is 1, and 0 otherwise.
- 在 $A \oplus B$ 的二进制表示中, 若 A 和 B 的二进制表示的 2^k 位恰好有一个是 1, 则该位为 1, 否则为 0。其中 2^k 位对应 $2^{\{k\}}$ 的位权。

For example, $3 \oplus 5 = 6$ (in binary: $011 \oplus 101 = 110$).

In general, the bitwise XOR of k non-negative integers $p_1, p_2, p_3, \dots, p_k$ is defined as $(\dots((p_1 \oplus p_2) \oplus p_3) \oplus \dots \oplus p_k)$, and it can be proved that this does not depend on the order of $p_1, p_2, p_3, \dots, p_k$.

例如, $3 \oplus 5 = 6$ (二进制表示为: $011 \oplus 101 = 110$)。一般地, k 个非负整数 $p_1, p_2, p_3, \dots, p_k$ 的按位 XOR 定义为 $(\dots((p_1 \oplus p_2) \oplus p_3) \oplus \dots \oplus p_k)$, 且可以证明该运算结果与 $p_1, p_2, p_3, \dots, p_k$ 的顺序无关。

Constraints

- $1 \leq T \leq 10$
- $1 \leq N, M$
- The sum of N over all test cases is at most 5×10^5 .
- The sum of M over all test cases is at most 5×10^7 .
- $1 \leq A_i \leq M$
- All input values are integers.
- $1 \leq T \leq 10$
- $1 \leq N, M$
- 所有测试数据的 N 之和不超过 5×10^5 。
- 所有测试数据的 M 之和不超过 5×10^7 。

- $1 \leq A_i \leq M$
 - 所有输入值均为整数。
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Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

```
T
case1
case2
⋮
caseT
```

Each test case is given in the following format:

每组测试数据按照以下格式给出：

```
N M
A1 A2 ... AN
```

Output

Output the answers for the test cases in order, separated by newlines.

按顺序输出每组测试数据的答案，相邻答案之间用换行分隔。

Sample Input 1

```
3
3 6
2 3 1
5 10
3 1 4 1 5
7 12
4 2 3 4 2 3 3
```

Sample Output 1

61
115
190

Consider the first test case.

我们来看第一组测试数据。

For example, when Takahashi's initial amount of money is 3 yen, he acts as follows.

例如，当高桥初始持有 3 日元时，他的行为如下：

- For $i = 1$: he has 3 yen, which is at least 2 yen, so he purchases the first item.
- For $i = 2$: he has 1 yen, which is less than 3 yen, so he does not purchase the second item.
- For $i = 3$: he has 1 yen, which is at least 1 yen, so he purchases the third item.
- 当 $i = 1$ 时：他有 3 日元，不少于 2 日元，因此购买第一件商品。
- 当 $i = 2$ 时：他有 1 日元，少于 3 日元，因此不购买第二件商品。
- 当 $i = 3$ 时：他有 1 日元，不少于 1 日元，因此购买第三件商品。

From this, we get $f(3) = 2 + 1 = 3$.

由此可得 $f(3) = 2 + 1 = 3$ 。

The values of $f(1), f(2), f(3), f(4), f(5), f(6)$ are 1, 2, 3, 3, 5, 6, respectively. Thus, the answer is the bitwise XOR of 1, 4, 9, 12, 25, 36, which is 61.

$f(1), f(2), f(3), f(4), f(5), f(6)$ 的值分别为 1, 2, 3, 3, 5, 6。因此答案是 1, 4, 9, 12, 25, 36 的按位 XOR，即 61。